ChatGPT

E120 Carmine and Cochineal Extract Evidence Registry Memo for Bari

Executive conclusion

E120 is a regulated red food color additive family that includes cochineal extract, carminic acid, and carmine. In official definitions, it is derived from the cochineal insect *Dactylopius coccus*; carminic acid is the main coloring principle, and carmine is a lake form built around that pigment. In practice, this makes E120 primarily an **ingredient transparency issue**, not a macronutrient or nutrient-quality issue. Bari should therefore **not reduce the nutrition score** for E120 on current evidence, but it **should** register E120 as a color additive that adds formulation complexity and deserves dietary-preference transparency because it is insect-derived. [1]

The safety signal is narrower than many consumers assume. EFSA's re-evaluation found carminic acid not genotoxic, found carmine not carcinogenic, and reported no adverse reproductive or developmental findings in the studies reviewed. EFSA retained the historic ADI of 5 mg/kg body weight per day and recommended expressing it as **2.5 mg carminic acid/kg bw/day**; in its refined non-brand-loyal exposure scenario, exposure was below the ADI for all population groups. JECFA likewise records a maintained ADI of **0–5 mg/kg bw for carmines**. [2]

The part that matters for user-facing interpretation is that **rare hypersensitivity and anaphylaxis are real but uncommon**, and the evidence is largely built from case reports, case series, and selected referral cohorts rather than high population-level incidence studies. FDA's final rule is explicit that cochineal extract and carmine are allergens for a **small subset** of the allergic population and are **not major food allergens** under FALCPA, which is why the right Bari posture is a **low-salience caution**, not a danger message. [3]

For Bari, the balanced rule is: **no nutrition penalty; modest additive-complexity increment; clear insect-derived transparency flag; non-alarmist hypersensitivity note; and careful label matching that avoids false positives such as "Cochineal Red A" and "Indigo Carmine," which are different additives.** [4]

Evidence table

Raw URLs are omitted below because the linked citations already open the source. For journal articles and regulations, the table uses DOI, PMID, INS number, or legal citation.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Scientific Opinion on the re-evaluation of cochineal, carminic acid, carmines (E 120) as a food additive	EFSA Journal	2015	DOI: 10.2903/j.efsa.2015.4288	EFSA concluded that carminic acid is not genotoxic, carmine is not carcinogenic, and no adverse reproductive or developmental effects were reported in the studies reviewed. EFSA retained the historic ADI and recommended expressing it as 2.5 mg carminic acid/kg bw/day; refined non-brand-loyal exposure was below the ADI for all population groups. EFSA also stated that the ADI does not cover allergic reactions because no threshold dose can be established for that hazard. [5]	High	Core source for Bari’s safety framing: regulated and generally safe at authorized use, but allergy is a separate rare-risk issue that should not be translated into a general danger signal.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
21 CFR 73.100 Cochineal extract; carmine	FDA / eCFR	current	21 CFR 73.100	FDA defines cochineal extract as the concentrated solution obtained from an aqueous-alcoholic extract of cochineal and states that carminic acid is the chief coloring principle. Carmine is defined as the aluminum or calcium-aluminum lake of the coloring principles obtained from cochineal. FDA allows use for coloring foods generally in amounts consistent with GMP, requires labeling by the common or usual name, and lists the additive as exempt from certification. [6]	High	Core identity source; supports insect-derived classification, by-name detection, and a complexity signal rather than a nutrition penalty.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Small Entity Compliance Guide: Declaration by Name on the Label of All Foods and Cosmetic Products that Contain Cochineal Extract and Carmine	FDA	2009, current page 2018	Guidance tied to 74 FR 207	FDA states that cochineal extract or carmine must be declared by name in the ingredient list of all human food products, including butter, cheese, and ice cream, and may not be hidden under generic color wording. [7]	High	Strong support for Bari label detection and for user-facing transparency.
Listing of Color Additives Exempt From Certification; Food, Drug, and Cosmetic Labeling: Cochineal Extract and Carmine Declaration	FDA / Federal Register	2009	74 FR 207	FDA states that cochineal extract and carmine are allergens for a small subset of the allergic population, are not major food allergens under FALCPA, and must nevertheless be labeled by name. FDA also explicitly declined to require mandatory “insect” or “animal-derived” wording on-pack. [8]	High	This is the best source for Bari’s non-alarmist allergy wording and for the transparency-gap logic around insect origin.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Carmines	WHO JECFA database	current; meetings 1982 and 2000 reflected	JECFA chemical 1079	JECFA records a maintained ADI of 0–5 mg/kg bw for carmines and concludes that cochineal extract, carmines, and possibly carminic acid may provoke allergic reactions in some individuals. [9]	High	Global regulatory anchor: safe within toxicological limits, with recognized but uncommon allergenicity.
Cochineal Extract and Carmines details pages	FAO JECFA	current	INS 120; CI 75470; CI Natural Red 4	FAO/JECFA lists INS number 120 and the synonyms CI Natural Red 4 and CI 75470 for cochineal extract and carmines. [10]	High	Supports synonym expansion for detection logic and registry aliases.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
GSFA Online Food Additive Details for Carmines	Codex / FAO	current	INS 120	Codex GSFA shows permitted use across multiple categories, including dairy-based desserts such as pudding and fruit or flavoured yoghurt, flavoured fluid milk drinks, confectionery, chewing gum, certain beverages, and some meat/fish applications such as glazes, coatings, surimi, and roe-related categories. [11]	High	Strong support for Bari use-case examples; also a reason to avoid overfitting E120 to only one product class.
Israeli adopted specifications for additives	Israeli Ministry of Health	2025 adopted publication	MOH adopted specs for additives	The Israeli adopted specifications list E120 as חומצה כרמינית / כרמין, give the synonym CI Natural Red 4, and specify composition and impurity limits. [12]	High	Best official Israeli source for Hebrew naming and local registry matching.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Israeli booklet on food additive labeling and Israeli food additive list	Israeli Ministry of Health	2015; 2022 publication	MOH additive-book; fcs_list	The booklet states that food additives appear under the “ingredients” heading on Israeli food labels. The additive list places E120 among permitted food colors in categories that include non-alcoholic beverages, baked goods, ice creams, desserts, and candies. [13]	High	Supports Hebrew label parsing and Israeli-market product examples.
The role of natural color additives in food allergy	Advances in Food and Nutrition Research	2001	PMID: 11285683; DOI: 10.1016/S1043-4526(01)43005-1	The review concluded that reactions to natural color additives are rare overall and that ingestion of natural color additives presents a very low risk of provoking adverse reactions, while also noting reported IgE-mediated adverse reactions to carmine. [14]	Medium	Good review-level context to keep Bari language conservative and non-alarmist.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Allergy to Carmine Red (E120) Is Not Dependent on Concurrent Mite Allergy	International Archives of Allergy and Immunology	2009	PMID: 19439984; DOI: 10.1159/000218121	In a selected group of patients with suspected additive allergy, 94 of 3,164 patients tested were skin-prick-test positive for carmine, or about 3.0%; this is a referral-population signal, not general-population prevalence. [15]	Medium	Useful for distinguishing sensitization in selected patients from population-wide risk.
An oral challenge test with carmine red (E120) in skin prick test positive patients	European Annals of Allergy and Clinical Immunology	2015	PMID: 26549338	In 23 symptomatic skin-prick-test-positive patients who underwent single-blind placebo-controlled oral challenge, 5 developed clinical symptoms. The authors estimated clinical carmine allergy at about 0.7% in general dermatology patients studied for food-associated symptoms. [16]	Medium	Stronger than case reports for Bari's caution language, but still not general-population incidence.

Source title	Organization or journal	Year	Locator	What it says	Evidence strength	Registry implication
Carmine allergy in urticaria patients	Advances in Dermatology and Allergology	2022	PMID: 35369613; DOI: 10.5114/ada.2020.100821	In a selected urticaria cohort of 110 patients with suspected additive hypersensitivity, carmine allergy was diagnosed in 9 patients, or 8%, after testing and challenge protocols. [17]	Medium	Supports caution for high-risk, symptomatic users; should not be generalized to everyone.
Cochineal dye-induced immediate allergy: Review of Japanese cases and proposed new diagnostic chart	Allergology International	2018	PMID: 29705083; DOI: 10.1016/j.alit.2018.02.012	The review states that severe allergic reactions can manifest as anaphylaxis and summarizes 22 Japanese cases. [18]	Medium	Good review support for “rare but real” wording.
Food anaphylaxis following ingestion of carmine	Annals of Allergy, Asthma & Immunology	1995	PMID: 7538438	A case report documented IgE-dependent anaphylaxis after yogurt ingestion and highlighted that a low dose could trigger symptoms in a sensitized individual. [19]	Low	Important only as a rare-case signal; should never be used to justify broad “danger” language.

Risk interpretation

Established facts

E120 is not one single consumer phrase but a regulatory family centered on cochineal extract, carminic acid, and carmine. Official sources consistently define it as an insect-derived red colorant from *Dactylopius* *coccus*, with carminic acid as the principal pigment; official synonym sets also include **CI Natural Red 4** and **CI 75470**, which matters for registry aliasing. [20]

It is used across a broad but not unlimited set of food categories. The strongest source-grounded use cases are flavored dairy desserts and yogurts, milk drinks, confectionery, chewing gum, some beverages, and certain meat/fish decoration or coating applications. I would **not** make plant-based meat or fish analogues a headline use case without better product-level evidence; Codex support is stronger for conventional meat/fish-related applications than for analogue products. [11]

Regulatory safety conclusions

The weight of regulatory evidence does **not** support treating E120 as generally hazardous at authorized uses. EFSA's 2015 re-evaluation found no genotoxic concern for carminic acid, no carcinogenic concern for carmine, and no adverse reproductive or developmental findings in the reviewed data. EFSA retained the ADI and translated it to **2.5 mg carminic acid/kg bw/day**; under refined non-brand-loyal exposure assumptions, exposure stayed below the ADI in all population groups. JECFA's database likewise maintains a **0–5 mg/kg bw** ADI for carmines. [2]

That said, EFSA draws an important boundary that Bari should preserve: the toxicological ADI is **not** an allergy threshold. EFSA states that no threshold dose can be established for allergic reactions in this context and advises minimizing exposure to eliciting allergens through purification-related controls. In other words, "safe within ADI" and "incapable of rare allergy" are not the same statement. [21]

Rare allergy and hypersensitivity evidence

The allergy signal is real, but the evidence base is mostly made of case reports, case series, review articles, and challenge studies in selected patient groups. FDA's rulemaking states that cochineal extract and carmine are allergens for a **small subset** of the allergic population and are not major food allergens. That is exactly why Bari should avoid heavy warning language. [8]

The clinical literature supports this narrower framing. Classic case reports describe IgE-mediated anaphylaxis after ingestion of carmine-containing foods, including yogurt and drinks. Review work from Japan summarized 22 reported cases and notes that severe reactions can include anaphylaxis. In selected clinical populations, the signal is higher: about 3% skin-prick-test positivity in one referred cohort, about 0.7% challenge-confirmed clinical allergy in general dermatology patients studied for food-associated symptoms, and 8% diagnosed allergy in a selected urticaria cohort. Those are **not** general-population prevalence estimates; they are selected-population findings and should be interpreted as such. [22]

A further nuance matters for explanation text: both JECFA and EFSA point toward allergenicity that is likely related to proteinaceous material or allergenic residues associated with commercial preparations, not a broad toxic effect shared by all consumers. This again argues for a **known-sensitivity caution**, not a universal avoidance recommendation. [23]

Consumer transparency and dietary-preference relevance

Bari should treat E120 as a transparency-sensitive additive because it is insect-derived. That matters straightforwardly for vegan and vegetarian users. It may also matter for kosher-sensitive and halal-sensitive users, but Bari should be careful here: the correct product-level determinant is still certification, not Bari guessing religious status from one ingredient alone. For kosher sensitivity specifically, an Israeli government kashrut guidance document states that carminic acid E120 is not kosher in its general guidance. For halal-sensitive users, Bari should signal insect origin and avoid a categorical halal ruling unless product certification is known. [24]

There is also a real transparency gap around “natural color” framing. FDA classifies cochineal extract and carmine as color additives exempt from certification and explicitly notes that cochineal extract is derived from an insect, but the agency requires disclosure by the names “cochineal extract” or “carmine,” not by a mandatory “insect-derived” statement. FDA also rejected mandatory insect-origin wording in the 2009 final rule. So consumer confusion is plausible, and Bari adding plain-language context is defensible. [25]

Unsupported or exaggerated claims to avoid

Bari should avoid saying or implying that E120 is “dangerous,” “toxic,” “carcinogenic,” or a reason for routine avoidance. That would overstate both the toxicology and the epidemiology. The regulatory consensus does not support that framing. [26]

Bari should also avoid calling E120 a “major allergen.” FDA is explicit that it is **not** a major food allergen under FALCPA, even though it can cause reactions in sensitive individuals. [8]

Finally, Bari should avoid two label-matching mistakes that are easy to make if the team relies on loose string logic. **Cochineal Red A** is **E124**, not E120, and **Indigo Carmine** is **E132**, not E120. If Bari matches those strings to the E120 registry entry, the model will generate false positives and lose trust. [27]

Bari Evidence Registry proposal

The operational mistake to avoid is treating E120 as either “nothing” because it is natural or “high risk” because it is insect-derived and occasionally allergenic. Both would be weak policy choices. The evidence supports a middle position: **nutrition-neutral, complexity-positive, transparency-forward, and low-salience allergy-aware.** [28]

signal_id	ingredient_names	Hebrew patterns	English patterns	additive_class	e
`additive_e120_carmine_presence`	E120; carmine; cochineal extract; carminic acid; CI Natural Red 4; CI 75470	כרמין; קרמין; חומצה כרמינית; צבע מאכל E120	E120; E-120; INS 120; carmine; cochineal extract; carminic acid; Natural Red 4; CI 75470	Color additive	H

signal_id	ingredient_names	Hebrew patterns	English patterns	additive_class	ev
`dietary_insect_derived_e120`	Same as above	כרמין; קרמין; קוציניל; קוצ'יניל; קוצ'יניל	cochineal; carmine; cochineal extract; carminic acid	Dietary-preference transparency	H o M p re in
`allergy_carmine_known_sensitivity_notice`	Same as above	כרמין; קרמין; חומצה כרמינית	carmine; cochineal extract; carminic acid	Hypersensitivity caution	M

Ownership

Owner: Chief Nutrition Officer
Why this owner: This is fundamentally a classification, wording, and evidence-calibration decision rather than a pure engineering task.

Copy-paste prompt

```
Approve Bari policy for E120/carmine/cochineal extract as follows: no nutrition-score penalty; modest additive-complexity increment; mandatory insect-derived transparency flag; low-salience caution for rare known hypersensitivity; no “dangerous/toxic” wording; no blanket avoidance recommendation; no categorical halal judgment without certification context.
```

Owner: Frontend Architect
Why this owner: Implementation quality depends on correct label matching, false-positive control, and rendering the right signal hierarchy.

Copy-paste prompt

```
Implement E120 detection with normalized matching for E120/E-120/INS 120/
carmine/cochineal extract/carminic acid/CI 75470/Natural Red 4 and Hebrew
variants כרמין/קרמין/חומצה כרמינית/קוציניל. Add explicit exclusions for
“Cochineal Red A” (E124) and “Indigo Carmine” (E132). Render three layers:
additive presence, insect-derived transparency, and low-salience
hypersensitivity notice.
```

Owner: QA & Audit Lead
Why this owner: The main product risk is misclassification, especially confusing E120 with E124 or E132 and over-warning users.

Copy-paste prompt

```
Build a regression pack for E120 with positives and negatives. Positives: E120,
E-120, INS 120, carmine, cochineal extract, carminic acid, CI 75470, Natural Red
4, כרמין, קרמין, חומצה כרמינית, קוצ׳יניל, קוצ'יניל. Negatives: Cochineal Red A
(E124), Indigo Carmine (E132), Carmoisine/Azorubine (E122), generic “color
added,” and unrelated red colorants.
```

Detection patterns

The safest implementation is a **two-step matcher**: first normalize text, then apply allow-list patterns plus a mandatory exclude-list. That matters because “carmine” appears in other additive names that are **not** E120, especially **Indigo Carmine** and **Cochineal Red A**. [27]

Recommended normalization before matching: 1. Uppercase Latin characters. 2. Normalize hyphens and dashes to a single `-`. 3. Normalize apostrophes and geresh variants to a single mark. 4. Collapse repeated whitespace. 5. Optionally strip punctuation around ingredient delimiters, but do **not** merge letters and numbers so that `E 120`, `E-120`, and `E120` remain detectable.

Pattern family	Suggested matching rule
E-number	`(?i)\bE[\s\-–—]?120\b`
INS number	`(?i)\bINS[\s\-–—]?120\b`
Core English names	`(?i)\bCARMINE\b` ; `(?i)\bCOCHINEAL(?:\s+EXTRACT)?\b` ; `(?i)\bCARMINIC\s+ACID\b`
CI and common synonyms	`(?i)\bCI[\s\-]?75470\b` ; `(?i)\b(?:CI\s+)?NATURAL\s+RED\s*4\b`
Core Hebrew names	`\bכרמין\b` ; `\bקרמין\b` ; `חומצה\s+כרמינית`
Hebrew consumer-market spellings	`קוציניל` ; `קוצ"יניל` ; `קוצ'יניל`

Pattern family	Suggested matching rule
Hebrew E-number phrase	`צבע\s*מאכל\s*E[\s\-]?120`

Recommended excludes, because they are **different additives**:

- `(?i)\bCOCHINEAL\s+RED\s+A\b` → this is **E124**, not E120. [27]
- `(?i)\bINDIGO\s+CARMINE\b` → this is **E132**, not E120. [27]
- `(?i)\bCARMOISINE\b|\bAZORUBINE\b` → this is **E122**, not E120. [27]

One additional product choice is worth challenging: do **not** match generic strings such as “natural color,” “colour,” or “color added” to E120. In the U.S., FDA specifically requires carmine/cochineal to be named by their common or usual names, which means a generic-color match is both legally weak and likely to create noise. [32]

Scoring recommendation

Question	Recommendation	Rationale
Should E120 reduce Bari nutrition score?	**No.**	The evidence base supports a regulatory-safe color additive framing, not a nutrient-quality downgrade. E120 does not worsen nutrient density in the way Bari’s nutrition score is usually meant to capture. EFSA and JECFA safety conclusions do not justify a nutrition penalty. [33]
Should it increase additive or UPF complexity?	**Yes, modestly.**	E120 is still a non-nutritive appearance-focused color additive. “Natural” should not exempt it from a formulation-complexity signal. The right move is a modest complexity increment, not a strong penalty. [34]
Should it trigger a dietary-preference transparency flag?	**Yes.**	It is insect-derived, which is directly relevant for vegan and vegetarian users and potentially relevant for kosher-sensitive or halal-sensitive users. Bari should surface origin clearly while avoiding overconfident religious-law judgments without certification context. [30]

Question	Recommendation	Rationale
Should it trigger an allergy or hypersensitivity caution?	**Yes, but low salience.**	FDA, JECFA, and the clinical literature support a “rare but real” hypersensitivity signal. That is enough for a caution note, especially useful for users with known sensitivities, but not enough for a general warning banner or avoidance recommendation. [35]
What wording is safe and non-alarmist?	**Recommended English:** “Contains E120 carmine, an insect-derived red color. Generally permitted for use; rare allergic reactions have been reported in sensitive individuals.” **Recommended Hebrew:** “מכיל כרמין E120, צבע אדום ממקור חרקי. מותר לשימוש רגולטורי; דווחו תגובות אלרגיות נדירות אצל אנשים רגישים.”	This wording is evidence-grounded, avoids calling the additive dangerous, and keeps the user interpretation aligned with FDA and EFSA framing. [36]

The key policy challenge is consistency. If Bari gives E120 a harsh risk treatment because it is insect-derived, Bari will be overstating the safety signal. If Bari gives it no signal because it is “natural,” Bari will be understating the transparency signal. The evidence supports a narrower and stronger policy: **nutrition-neutral, modestly complexity-positive, clearly insect-derived, and quietly cautionary for rare known hypersensitivity.** [37]

---

[1] [6] [20] [24] [30] [34] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-73/subpart-A/section-73.100
https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-73/subpart-A/section-73.100

[2] [5] [26] [28] [29] [33] [37] https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2015.4288
https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2015.4288

[3] [8] [31] [35] [36] https://www.federalregister.gov/documents/2009/01/05/E8-31253/listing-of-color-additives-exempt-from-certification-food-drug-and-cosmetic-labeling-cochineal
https://www.federalregister.gov/documents/2009/01/05/E8-31253/listing-of-color-additives-exempt-from-certification-food-drug-and-cosmetic-labeling-cochineal

[4] [12] [27] https://mohpublic.health.gov.il/FCS/231_2012_specifications_for_additives_in_13332008_HE.pdf
https://mohpublic.health.gov.il/FCS/231_2012_specifications_for_additives_in_13332008_HE.pdf

[7] [32] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/small-entity-compliance-guide-declaration-name-label-all-foods-and-cosmetic-products-contain
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/small-entity-compliance-guide-declaration-name-label-all-foods-and-cosmetic-products-contain

[9] [23] https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/1079
https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/1079

[10] https://www.fao.org/food/food-safety-quality/scientific-advice/jecfa/jecfa-additives/details-simple/en/?ins_id=120

https://www.fao.org/food/food-safety-quality/scientific-advice/jecfa/jecfa-additives/details-simple/en/?ins_id=120

[11] https://www.fao.org/gsfaonline/additives/details.html%3Bjsessionid%3D49E532AC4D0DA6AA0FF1E2C2778CCC38?d-3586470-o=2&d-3586470-s=4&id=89

https://www.fao.org/gsfaonline/additives/details.html%3Bjsessionid%3D49E532AC4D0DA6AA0FF1E2C2778CCC38?d-3586470-o=2&d-3586470-s=4&id=89

[13] https://www.gov.il/BlobFolder/reports/additive-book/he/files_publications_units_public_health_services_additive_book.pdf

https://www.gov.il/BlobFolder/reports/additive-book/he/files_publications_units_public_health_services_additive_book.pdf

[14] https://pubmed.ncbi.nlm.nih.gov/11285683/

https://pubmed.ncbi.nlm.nih.gov/11285683/

[15] https://karger.com/iaa/article/150/2/179/165613/Allergy-to-Carmine-Red-E120-Is-Not-Dependent-on

https://karger.com/iaa/article/150/2/179/165613/Allergy-to-Carmine-Red-E120-Is-Not-Dependent-on

[16] https://pubmed.ncbi.nlm.nih.gov/26549338/

https://pubmed.ncbi.nlm.nih.gov/26549338/

[17] https://www.termedia.pl/Carmine-allergy-in-urticaria-patients%2C7%2C42400%2C0%2C1.html

https://www.termedia.pl/Carmine-allergy-in-urticaria-patients%2C7%2C42400%2C0%2C1.html

[18] https://pubmed.ncbi.nlm.nih.gov/29705083/

https://pubmed.ncbi.nlm.nih.gov/29705083/

[19] [22] https://pubmed.ncbi.nlm.nih.gov/7538438/

https://pubmed.ncbi.nlm.nih.gov/7538438/

[21] https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2015.4288

https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2015.4288

[25] https://www.fda.gov/food/color-additives-information-consumers/color-additives-foods

https://www.fda.gov/food/color-additives-information-consumers/color-additives-foods